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Privacy Preserving Inference for Deep Neural Networks: Optimizing Homomorphic Encryption for Efficient and Secure Classification

Muhammad Aftab Akram, [Fawad Khan](#), Shahzaib Tahir, Asif Iqbal, [Syed Aziz Shah](#), Abdullah Baz

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Abstract

The application of machine learning in healthcare, financial, social media, and other sensitive sectors not only involves high accuracy but privacy as well. Due to the emergence of the Cloud as a computation and one-to-many access paradigm; training and classification/inference tasks have been outsourced to Cloud. However, its usage is limited due to legal and ethical constraints regarding privacy. In this work, we propose a privacy-preserving neural networks-based classification model based on Homomorphic Encryption (HE) where the user can send an encrypted instance to the cloud and receive an encrypted inference from it to preserve the user's query privacy. In contrast to existing works, we demonstrate the realistic limitations of HE for privacy-preserving machine learning by changing its parameters for enhanced security and accuracy. We showcase scenarios where the choice of HE parameters impedes accurate classification and present an optimized setting for achieving reliable classification. We present several results to demonstrate its effectiveness using MNIST dataset with highly improved inference time for a query as compared to the state of the art.

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Keywords

- Convolutional neural network
- homomorphic encryption
- activation function
- cloud server
- approximation techniques

- security and privacy
- encrypted computations

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KW - activation function

KW - cloud server

KW - approximation techniques

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